UNIT 1: UNITS AND MEASUREMENTS

A quick, color-coded rewrite of the handwritten notes

Physical quantities & measurement

Physics is a quantitative science: we measure physical properties in experiments and everyday life. A measurement compares a quantity with an accepted standard unit.

Measurement = numerical value × unit
Example: 5 m means a length with magnitude 5, measured in metres.

Units

A unit is the standard used to measure a physical quantity. Different quantities use different units.

Fundamental quantities

Fundamental quantities do not depend on other physical quantities for their measurement. The seven SI base quantities are:

Quantity	SI unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Electric current	ampere	A
Luminous intensity	candela	cd
Amount of substance	mole	mol

Derived quantities & units

A derived quantity depends on one or more other physical quantities. A derived unit is formed from base units.

Quantity	SI unit	Symbol
Area	square metre	m^2
Volume	cubic metre	m^3
Density	kilogram per cubic metre	kg/m^3
Velocity	metre per second	m/s
Force	newton	$N = kg \cdot m/s^2$
Energy / work	joule	J
Power	watt	W

Supplementary quantities in the notes

Quantity	SI unit	Symbol
Plane angle	radian	rad
Solid angle	steradian	sr

Common systems of units

- CGS: centimetre, gram, second
- FPS: foot, pound, second
- MKS: metre, kilogram, second
- SI: International System of Units; the standard system used for science and measurement

Scalar vs vector

Type	What describes it?	Examples
Scalar	Magnitude only	length, mass, time, temperature, speed, energy
Vector	Magnitude and direction	displacement, velocity, acceleration, force, weight

Dimensions & dimensional formula

The dimensions of a quantity show the powers of base quantities needed to express it. A dimensional formula writes a derived quantity using base symbols in square brackets.

Base dimensions: length [L] • mass [M] • time [T] • temperature [Θ] • current [I] • luminous intensity [J] • amount [N]

The SI base units are m, kg, s, K, A, cd and mol. Dimensional symbols: L, M, T, Θ, I, J and N.

Quantity	SI unit	Dimensional formula
Area	m^2	$[L^2]$
Volume	m^3	$[L^3]$
Density	kg/m^3	$[M\ L^{-3}]$
Velocity	m/s	$[L\ T^{-1}]$
Acceleration	m/s^2	$[L\ T^{-2}]$
Force	N	$[M\ L\ T^{-2}]$
Momentum	$kg·m/s$	$[M\ L\ T^{-1}]$
Energy / work	J	$[M\ L^2\ T^{-2}]$
Power	W	$[M\ L^2\ T^{-3}]$
Pressure	N/m^2	$[M\ L^{-1}\ T^{-2}]$

Errors in measurement

Measurement results have uncertainty. The notes group errors into two broad types:

- Systematic errors: caused by a consistent problem in the instrument, method, setup or observer. Examples include a faulty meter, a consistent reading mistake, parallax, or a graduated tube held at an angle.
- Random errors: unpredictable variations that remain after careful procedure. They cannot be removed completely; repeated readings and averaging help reduce their effect.

Reduce systematic error: use a suitable, correctly working instrument; follow the correct procedure; read scales carefully and avoid parallax.

Estimating error from repeated readings

For observations $a_1, a_2, \ldots, a_n$:

Mean: $a_{mean} = (a_1 + a_2 + \ldots + a_n) / n$

Absolute error for each reading: $|\Delta a_i| = |a_i - a_{mean}|$

Mean absolute error: $\Delta a_{mean} = (|\Delta a_1| + |\Delta a_2| + \ldots + |\Delta a_n|) / n$

Relative error = $\Delta a_{mean} / a_{mean}$

Percentage error = $(\Delta a_{mean} / a_{mean}) \times 100\%$

Worked error example

Readings: 1.1, 1.2 and 1.3

Step	Calculation	Result
Mean	(1.1 + 1.2 + 1.3) / 3	1.2
Absolute errors	$\lvert 1.2-1.1 \rvert$, $\lvert 1.2-1.2 \rvert$, $\lvert 1.2-1.3 \rvert$	0.1, 0, 0.1
Mean absolute error	(0.1 + 0 + 0.1) / 3	0.067
Relative error	0.067 / 1.2	≈ 0.056
Percentage error	0.056 × 100%	≈ 5.6%

A measurement written as 50 ± 2 cm has percentage error = (2 / 50) × 100% = 4%.

Significant figures

Significant figures are the meaningful digits in a measured number.

- All non-zero digits are significant. Example: 347 has 3 significant figures.
- Zeros between non-zero digits are significant. Example: 2004 has 4.
- Trailing zeros after a decimal point are significant. Example: 2.1000 has 5.
- Trailing zeros in a whole number without a decimal point are not assumed significant. Example: 100000 may show only 1 significant figure.
- Leading zeros in a number below 1 are not significant. Example: 0.0012 has 2.
- In scientific notation, count the digits in the coefficient; the power of ten does not add significant figures. Examples: $1 \times 10^{\square}$ has 1; 6.2×10^{3} has 2.

Measurements in daily life

- Medical checks: temperature, pulse and blood counts.
- Home appliances: temperature and time settings help cook or heat food.
- Transport: fuel, air pressure, oil and speed are measured.
- Shopping: food, groceries, vegetables and clothing are measured.

Measurements are part of everyday decisions, science and technology.

Ancient astronomical instruments

The notes describe instruments used in India to estimate large distances and heights, measure angles, track star positions and tell time. Bhaskaracharya (Bhaskara II), an Indian mathematician and astronomer, designed instruments and wrote Siddhanta Shiromani.

Instrument	What the notes say it was used for
Chakra yantra (wheel)	A disc-like protractor for marking angles and locating celestial positions; also used to estimate time from shadows and altitude.
Dhanur yantra (bow)	A semicircular instrument for vertical angles and estimating heights, Earth / Moon dimensions and distances.
Yashti yantra (staff)	A cross-staff used to calculate the height of large objects such as trees or hills.
Phalaka yantra (board)	A board used to find time from the Sun's altitude.

End of Unit 1 notes